

Thesis for Bachelor's Degree

Revisiting Bitcoin Price Prediction with On-Chain
and Financial Data: A Reflection on Reproducibility

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College of Information and Computing

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학 사 학 위 논 문

온체인 및 금융 데이터를 활용한 비트코인 가격
예측의 재검토: 재현성에 대한 고찰

한영서

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Revisiting Bitcoin Price Prediction with On-Chain and Financial Data: A Reflection on Reproducibility

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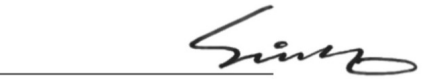
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Dedicated to my family.

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Abstract

Abstract— With recent advancements in machine learning and deep learning, algorithmic trading has increasingly leveraged sophisticated time-series prediction models [1]. As Bitcoin’s price continues to rise, interest has grown not only in its price dynamics but also in the potential societal impacts [2]. In this domain, methods based on LSTM, GRU, and Transformer architectures have been predominantly employed [1, 3]. Many studies claim superior predictive accuracy by combining extensive data collection, preprocessing, and feature transformation techniques. This study aims to assess whether the high accuracy reported in previous research—based on time-series data for predicting future prices—remains valid under more realistic conditions. By examining the performance of conventional models trained on both on-chain and market data [2], we seek to offer insights into more reliable asset price forecasting.

In our research, we generate explanatory variables for future prices by integrating not only basic time-series values but also various financial technical indicators. The objective is to predict asset prices and their directional movements over a given period,

with labels such as “Hold,” “Buy,” and “Sell” assigned based on specific thresholds; the corresponding investment performance is then evaluated to discuss the model’s rationality [4].

Experimental results indicate that the Transformer consistently outperforms conventional LSTM and GRU methods—particularly when both on-chain and market data are utilized—producing a more pronounced improvement in predictive accuracy [1]. The Transformer not only demonstrates higher accuracy in on-chain data analysis but also excels in key financial metrics, maintaining fundamentally better performance even after hyperparameter tuning. Furthermore, an examination of the maximum and minimum accuracies attained through variations in random seed values and parameter settings reveals that realistic accuracy tends to be lower than previously reported [3]. These findings underscore that Transformer-based asset price predictions are indeed meaningful for real-world applications, while also emphasizing the need for a cautious interpretation of claims regarding excessively high accuracy.

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국 문 요 약

머신러닝 및 딥러닝의 발전에 따라 알고리즘 트레이딩 분야에서는 이를 활용한 연구가 활발히 이루어지고 있다. 비트코인의 가격이 증가함에 따라 가격 역학에 따른 사회적 변화에 대한 관심 또한 커지고 있다 [2]. 이 분야에서는 LSTM, GRU, 트랜스포머 (Transformers) 기반 기법들이 대부분의 연구에 사용되어지고 있다 [1]. 대부분의 연구에서는 데이터 수집, 프리프로세싱, 여러 기법의 합성, 기법들의 변환을 통해서 자신들의 정확도나 예측 정도의 우월성을 주장한다. 본 연구에서는 시계열 데이터를 활용하여 미래의 가격을 예측하는 기존의 연구성이 주장하는 정확도가 타당한가에 대해 분석을 진행하고자 한다. 온체인 및 시장 데이터들을 활용하여 전통적인 모델로 학습하였을 때의 결과를 바탕으로 분석을 진행하고, 합리적인 미래 가격 예측에 대한 통찰을 제공하고자 한다.

본 연구에서는 시계열 데이터의 기초적인 값뿐만 아니라 금융 기술 지표(Technical Indicators)들을 활용하여 미래의 가격을 설명할 수 있는 변수들을 만든다. 일정 기간 동안 연속된 데이터의 정보를 기반으로 자산 가격과 방향성을 예측하는 것이 목표이다. 특정 임계값을 기준으로 Hold, Buy, Sell 등의 라벨링을 거치고 각각의 라벨에 따라

투자를 했을 때의 성과도 도출하여 미래 예측 모델의 합리성에 대해서 논한다 [4].

실험 결과, Transformer는 기존의 LSTM, GRU 기법보다 일관되게 더 나은 성능을 보였으며, 온체인 시장 데이터를 활용했을 때 그 차이가 두드러졌다 [1]. 온체인 데이터를 활용한 분석 기법에서의 정확성뿐만 아니라 금융 평가 지표에서도 뛰어난 성능을 보였으며, 하이퍼파라미터 튜닝에 대해서도 기본적으로 더 뛰어난 성능을 보였다. 랜덤 시드 값의 변환, 파라미터 튜닝에 대한 최고 최저 정확도를 분석함으로써 합리적인 정확도는 기존 연구들에 비해 비교적 낮은 값을 얻을 수 있었다 [3]. 이러한 연구 결과는 트랜스포머 기반 자산 가격 및 방향성 예측이 실제 상황에서 유의미하다는 것을 보여주며, 또한 과학계 높은 정확도에 대한 고찰이 필요함을 시사한다.

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List of Algorithms

Chapter 1

Introduction

1.1 Introduction

The prediction of asset prices (including equities) using artificial intelligence systems has been under investigation for nearly three decades. Indeed, price remains a central question of great interest to those engaged in financial activities. The financial ecosystem can be broadly observed from three perspectives: financial products, the individuals who trade these products, and the vast amounts of data generated by their transactions. In contemporary markets, tradeable instruments extend beyond equities and now include options, exchange-traded funds (ETFs), cryptocurrencies, commodities, and NFTs. Similarly, the types of participants in modern financial markets vary widely in approach—ranging from scalping, day trading, swing trading, and long-term investing to news-based trading, quantitative strategies, value investing, and macro-oriented strategies. As these participants trade financial instruments, prices change accordingly, generating multiple data streams such as market prices and volumes. These fluctuations can be observed through financial technical indicators or on-chain data. As the volume of data grows exponentially, it becomes increasingly challenging for individuals to process all of it comprehensively. Consequently, the role of AI-based trading systems has gained even greater importance across diverse global markets.

Although traditional Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) models have been employed for financial forecasting, their adoption in finance appears less advanced compared to other AI domains such as natural language processing (NLP). Transformers, originally developed for sequence analysis in NLP, have shown superior performance in various time-series analysis tasks and have surpassed traditional models in certain areas. By leveraging attention and self-attention mechanisms, Transformers can effectively learn relationships within sequences (or images). In particular, self-attention excels at capturing long-range dependencies by learning relationships among multiple components of the input.

An important question arises, however, regarding the reliability of accuracy claims made by previous studies. Before the advent of NLP-based models, a number of papers reported high accuracy using traditional approaches. Even now, researchers assert that state-of-the-art models like Transformers achieve higher accuracy than conventional methods. Yet, it remains uncertain whether such accuracy levels—often exceeding 70

In this study, we aim to provide a rational forecast of asset prices and their fluctuations, leveraging both well-established and currently popular techniques within an algorithmic trading framework. Three traditional methods—LSTM, and GRU—were employed, alongside a Transformer-based approach adapted from NLP. Specifically, we utilize algorithms that incorporate attention and self-attention mechanisms from Transformer architectures.

We incorporate 19 diverse technical indicators to capture a multidimensional representation of financial time-series data, directly using them as input features to our

models. These indicators reflect market volatility across specific periods and include critical measures such as MACD and RSI for trend and momentum analysis, as well as Bollinger Bands and ATR for volatility measurement. This approach enables a more effective modeling of the dynamic characteristics present in financial markets.

LSTM Model: This RNN-based model is designed to learn long-term dependencies, employing 25 LSTM units, and is structured to predict one of three classes (Buy, Hold, Sell) in its output layer.

GRU Model: While similar in structure to LSTM, it offers a simplified memory cell design for faster learning and robust performance even with relatively small datasets. In this research, 64 GRU units were utilized, followed by an additional 32 GRU layers to optimize performance.

Transformer-Based Model: Building on the original Transformer architecture, this model adopts Multi-Head Attention (MHA) and Layer Normalization, making it better suited for extracting long-term patterns compared to LSTM and GRU. For this study, two Transformer Encoder Blocks were used to process the input sequences, after which Global Average Pooling was applied to generate the final output.

By employing these models, the primary goal of this research is not merely to maximize accuracy but to explore whether the models can be used feasibly in real-world settings. In doing so, we aim to provide insights into more reasonable approaches for future price prediction, going beyond the pursuit of exceptionally high accuracy scores.



Figure 1.1: The symbol of GIST.

1.2 Related Work

In this study, we aim to forecast both the volatility and direction of asset prices by leveraging a multidimensional time series dataset that encompasses various financial assets as well as Bitcoin’s on-chain data. The dataset includes equity and ETF (Exchange-Traded Fund) price data, Bitcoin on-chain data, macroeconomic indicators, and market sentiment data, each of which can influence asset prices in significant ways.

1.2.1 Data Composition

The data employed in this research comprises key ETF and Bitcoin price information (OHLCV). Specifically, the assets include:

- BTC (Bitcoin)
- DIA (SPDR Dow Jones Industrial Average ETF)
- EEM (iShares MSCI Emerging Markets ETF)
- EFA (iShares MSCI EAFE ETF)
- IAU (iShares Gold Trust)

- IWM (iShares Russell 2000 ETF)
- QQQ (Invesco QQQ Trust, Nasdaq-100 ETF)
- SPY (SPDR S&P 500 ETF Trust)
- XLE (Energy Select Sector SPDR Fund)
- XLF (Financial Select Sector SPDR Fund)
- XLK (Technology Select Sector SPDR Fund)

These datasets provide OHLCV variables (Open, High, Low, Close, Adjusted Close, and Volume), which are essential for capturing and analyzing the volatility of financial assets. Building upon the OHLCV data, we generate numerous *technical indicators* to evaluate market volatility and price movements. The primary technical indicators employed in this study are as follows:

- **CCI (Commodity Channel Index):** Detects overbought or oversold conditions by measuring the mean deviation of price.
- **RSI (Relative Strength Index):** Assesses the relative strength of price by comparing the magnitude of recent gains to recent losses.
- **MACD (Moving Average Convergence Divergence):** Captures trend reversals by examining the difference between short-term and long-term moving averages.
- **Bollinger Bands:** Analyzes price volatility based on standard deviation around a moving average.

- **ATR (Average True Range):** Serves as a volatility indicator, measuring the average price fluctuation over a specified period.
- **SMA (Simple Moving Average) and EMA (Exponential Moving Average):** Reflect short- and long-term trends using moving averages.
- **OBV (On-Balance Volume):** Evaluates buying and selling pressure by considering volume changes.

These technical indicators help analyze the movement of individual financial assets and facilitate more granular modeling of market volatility.

1.2.2 On-Chain Data

On-chain data refers to information directly extracted from a blockchain network, encompassing metrics such as Bitcoin network activity, mining data, and transaction patterns. It serves as a vital tool for understanding market sentiment and price dynamics. The primary on-chain metrics used in this study include:

- **BTC / Difficulty (Bitcoin network difficulty):** Reflects the level of competition and security in mining. When difficulty increases, mining participation generally grows, potentially leading to a reduced supply of newly mined coins in the long term, thus influencing price increases.
- **BTC / Tx Cnt (Bitcoin transaction count):** Denotes the total number of transactions on the blockchain over a given period. Higher transaction counts

indicate more active participation on the network, which can be interpreted as a bullish signal for the market.

- **BTC Mean Tx Fee (Bitcoin average transaction fee, in USD):** Represents the average fee required for transactions within the network. Surges in transaction fees may point to higher network congestion, and could be associated with short-term price spikes or corrections.

Because Bitcoin’s price is closely linked to these on-chain metrics, they serve as key indicators of market participants’ behavior. For instance, an increase in transaction count (Tx Cnt) suggests heightened user activity, which can be seen as a bullish indicator, while rising mining difficulty (Difficulty) often reduces selling pressure and may bolster upward price movements. By integrating on-chain data with conventional financial (OHLCV) information, researchers can construct more precise models for price forecasting.

1.2.3 Traditional Financial Time-Series Forecasting Models

Early research on asset-price forecasting primarily relied on statistical time-series models. A representative example is the ARIMA (Autoregressive Integrated Moving Average) model, which is widely employed for time-series data analysis and forecasting in finance. While ARIMA proves useful for projecting future prices based on historical data, it has notable drawbacks in handling non-linearity and long-term dependencies.

Subsequently, GARCH (Generalized Autoregressive Conditional Heteroskedasticity) models were introduced to address market volatility, leading to studies on the

volatility of financial time series. Nonetheless, these statistical methods have limitations in capturing the full complexity of market fluctuations and in accommodating high-dimensional financial data.

To overcome such constraints, machine learning and deep learning techniques began to be applied to financial forecasting. Classical supervised learning models such as Random Forest and XGBoost were utilized to incorporate multiple variables (technical indicators) with the aim of improving predictive performance. However, they often failed to adequately account for the time-dependent properties inherent in financial markets, which prompted the adoption of Recurrent Neural Networks (RNNs).

1.2.4 LSTM and GRU-Based Financial Forecasting

To tackle the limitations of traditional machine learning in finance, deep learning models specialized in time-series analysis have been actively studied. In particular, LSTM (Long Short-Term Memory) and GRU (Gated Recurrent Unit), which are extensions of RNNs capable of learning long-term dependencies, have been widely applied to financial forecasting.

LSTM excels at modeling the continuous patterns in financial markets by effectively capturing both long-term and short-term memory, whereas GRU features a simplified gating structure that reduces parameter count and improves computational efficiency, enabling faster training. Numerous studies utilizing LSTM and GRU have analyzed price trajectories and trading signals in various markets, demonstrating superior performance to traditional ARIMA and GARCH models.

Despite their success, these models face several challenges:

- Strength in learning short-term patterns, but less effective in capturing longer-horizon market shifts
- Limited parallelization capabilities, resulting in slower training speeds
- Vulnerability to overfitting in highly volatile markets

1.2.5 Transformer-Based Financial Forecasting

The Transformer architecture, originally devised for sequence analysis in Natural Language Processing (NLP), has recently been introduced to the domain of time-series forecasting and financial prediction. Its core strength lies in the attention and self-attention mechanisms, which facilitate more effective learning of long-range dependencies compared to LSTM and GRU. Additionally, Transformers support parallel processing for more efficient training.

Recent work has proposed specialized Transformer variants for financial time-series data, such as Financial Transformer (FinBERT) and Time-Series Transformer (TST), suggesting that these models may replace traditional RNN-based architectures. Other studies have transformed financial time-series data into 2D images to employ Vision Transformers (ViT) or Swin Transformers for forecasting. Nevertheless, questions remain about the practical usage of Transformer-based models in real-world financial scenarios. In particular, doubts persist regarding whether deep learning models that achieve over 70% accuracy under certain conditions can consistently replicate such performance across diverse market environments. These concerns highlight the potential

for inflated accuracy results if models are tested only on specific market subsets.

To address these issues, the present research compares traditional models (LSTM, GRU) with state-of-the-art Transformer approaches, aiming to evaluate the practicality and robustness of each method in actual financial markets. By considering both on-chain and classical OHLCV data, this study seeks to identify a more realistic baseline for predictive performance, paving the way for more reliable and interpretable financial forecasting.

The Transformer model used in this study was implemented using TensorFlow 2.11 with the Keras API. Specifically, the ‘MultiHeadAttention’ and ‘LayerNormalization’ layers were employed to construct two encoder blocks, followed by a ‘GlobalAveragePooling1D’ layer and a classification head. While this implementation is based on the original ”Attention Is All You Need” architecture [Vaswani et al., 2017][5], position encodings were omitted, as the model was trained on ordered sequences assuming implicit temporal encoding.

The full implementation details are available at: https://github.com/kangwoul2/Graduation_Paper.git [6]

1.3 Data Preprocessing

1.3.1 Dataset Preparation

Financial data are generally analyzed through *technical analysis (TA)*, *fundamental analysis*, or *quantitative analysis*. While fundamental analysis evaluates a firm’s long-term investment value based on its financial statements, technical analysis focuses on

predicting market movements using price and volume data.

In this study, we perform time-series analysis by combining technical analysis with on-chain data. Data were collected using the **yfinance** library, targeting the following assets:

- Bitcoin (BTC)
- Exchange-Traded Funds (ETF):
 - SPY (S&P 500 ETF), QQQ (Nasdaq-100 ETF), DIA (Dow Jones ETF)
 - XLE (Energy ETF), XLF (Financial ETF), XLK (Technology ETF)
 - IWM (Russell 2000 ETF), EEM (Emerging Markets ETF), EFA (EAFE ETF), IAU (Gold ETF)

Each dataset includes **OHLCV data** (Open, High, Low, Close, Adjusted Close, and Volume), which were utilized to analyze price volatility in financial markets.

1. Analysis Period The analysis period in this study spans approximately five years, from January 1, 2018 to January 20, 2023, encompassing major financial market shocks such as the Luna incident in 2022 and the COVID-19 pandemic in 2020. By covering this timespan, we assess how extreme volatility and market instability influence model performance.

2. Technical Indicators To reflect market volatility, we incorporate various *technical analysis indicators*. Each indicator serves functions such as trend analysis, volatility

measurement, momentum evaluation, and volume analysis. The main indicators employed are as follows:

- **Trend-Based Indicators:** MACD, SMA(50-day), EMA(20-day), WMA(20-day), TEMA(20-day)
- **Momentum-Based Indicators:** RSI(14-day), CCI(14-day), ROC, WILLR(14-day)
- **Volatility-Based Indicators:** ATR(14-day), Bollinger Bands (Upper/Middle/Lower)
- **Volume-Based Indicators:** OBV, AD, Chaikin AD

These indicators were computed using Python’s **TA-Lib** (Technical Analysis Library) to model the multidimensional volatility of financial markets.

1.3.2 On-Chain Data Preparation

On-chain data are extracted directly from the blockchain network and can serve as critical variables for analyzing Bitcoin network activity and market sentiment. In this study, we included the following key on-chain metrics:

- **BTC / Difficulty (Bitcoin Network Difficulty):** Reflects mining difficulty, which indicates the level of network security and mining competition. Increasing difficulty typically implies a rise in mining participants, potentially leading to a reduced supply over the long term, which can be associated with upward price movement.

- **BTC / Tx Cnt (Bitcoin Transaction Count):** Denotes the total number of transactions occurring on the blockchain over a given period. A higher transaction count indicates increased network activity, which may correlate with greater investor interest and thus influence price.
- **BTC Mean Tx Fee (Average Bitcoin Transaction Fee, USD):** Represents the average fee needed to process transactions on the network. A spike in transaction fees can signal network congestion, potentially causing short-term price volatility and impacting Bitcoin’s price.

Additionally, we incorporate macroeconomic indicators such as CCI30 (cryptocurrency market index), the Fear & Greed Index, the S&P 500 Index, 10-year Treasury yields, gold prices, and the Consumer Price Index (CPI) to enhance the multidimensionality of our data. On-chain data contribute to a more nuanced understanding of Bitcoin’s supply and demand dynamics and, when combined with traditional market data, can help build more sophisticated predictive models.

1.3.3 Label Generation

To conduct a supervised learning-based prediction, we convert Bitcoin’s price movements into three classes: Buy, Hold, and Sell.

1. Threshold Definition Each sample is labeled according to a *predefined threshold value*:

- If the price change exceeds the threshold, assign a **Buy** label.

- If the price change is below the threshold, assign a **Sell** label.
- Otherwise, assign a **Hold** label.

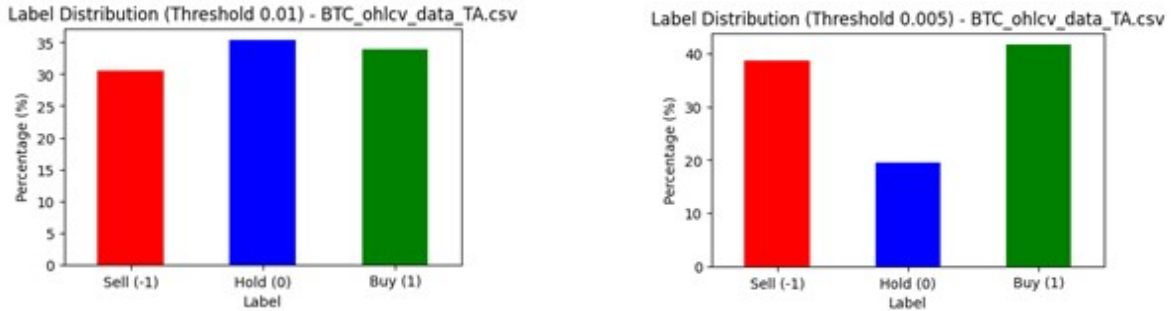


Figure 1.2: Comparison of data distribution with threshold 0.01 (left) and 0.005 (right).

We compared two threshold values, 0.01 and 0.005, and found that a threshold of 0.01 yielded a nearly balanced distribution among Buy, Hold, and Sell classes. Hence, a final threshold of 0.01 was adopted in training.

1.3.4 Data Normalization and Time-Series Transformation

1. Data Normalization Since each technical indicator and on-chain metric may have a different unit or scale, we applied the `StandardScaler` to normalize the data. This step ensures stable model training and prevents any single variable from disproportionately influencing the learning process.

2. Time-Series Transformation In this study, we varied the *lookback window* from 10 to 100 days to explore different spans of historical data:

- **X (Input):** technical indicators + on-chain data + macroeconomic data
- **y (Target):** Bitcoin price data

- **Classification:** Predicting whether Bitcoin’s price will move up, down, or remain flat ($\pm 0.5\%$)
- **Regression:** Predicting the next day’s rate of price change

The time-series dataset was transformed into overlapping sequences using a sliding window approach, with each window representing one training sample of 10 to 100 days of historical data. Each timestep (token) within a sequence includes 19 technical indicators, 3 on-chain variables, and selected macroeconomic features, resulting in a multidimensional vector. These sequences are treated as "token sequences" in the Transformer model, analogous to word tokens in NLP. We split the dataset into an 80:20 ratio for training and testing, maintaining the chronological order of the data to prevent leakage and to more accurately reflect real market conditions.

1.3.5 Data Organization and Model Input Format

The final dataset used in this research encompasses:

- **Financial Data (OHLCV)**
- **On-Chain Data** (BTC Difficulty, Tx Cnt, Mean Tx Fee, etc.)
- **Macroeconomic Data** (S&P 500, 10-year Treasury yields, gold prices, CPI, etc.)
- **19 Technical Indicators** (RSI, MACD, ATR, etc.)
- **Lookback Window:** 10–100 days

- **Target Variable:** Classification (Up/Down) or price rate-of-change (Regression)

These indicators are computed over various time horizons ranging from one to three weeks. Using this dataset, we compare traditional financial time-series forecasting models (LSTM, GRU) with a Transformer-based model. Notably, these technical indicators are optimized for *short- and medium-horizon trading*, which is the focus of this study. Should the scope shift to *long-term trading*, the selection and parameters of these technical indicators may require adjustment.

Chapter 2

Methodology

2.1 Data Preparation

This study utilizes a time-series dataset spanning from January 1, 2018, to January 20, 2023, for predicting the prices of financial assets (Bitcoin and major Exchange-Traded Funds). This timeframe encompasses extreme market conditions, including the COVID-19 pandemic (2020) and the Luna collapse (2022), making it well-suited for evaluating how models generalize under diverse market scenarios. The financial data were collected in the form of OHLCV (Open, High, Low, Close, Adjusted Close, Volume) and further supplemented with on-chain metrics (e.g., Bitcoin network difficulty, transaction fees) and macroeconomic indicators (e.g., interest rates, market sentiment indices). After addressing missing values and outliers in the dataset, a standard scaling procedure (`StandardScaler`) was applied to unify different units and scales.

The full implementation code used in this study, including data preprocessing, model training scripts, and experimental setup, is publicly available at:

`**https://github.com/your-username/bitcoin-transformer** [6]`

This repository enables full reproducibility of the experiments described in this thesis.

2.2 Methodology and Experimental Design

2.2.1 Model Selection and Experiment Setup

This study compares four types of forecasting approaches for predicting future price movements (*Buy, Sell, Hold*): two recurrent neural network (RNN) models (LSTM, GRU), and a Transformer-based model. All models were subject to the same data preprocessing and labeling schemes (i.e., threshold criteria, number of epochs, and optimizer settings), so as to ensure experimental consistency across architectures.

2.2.2 Two Core Experiments

We conducted two main experiments under carefully controlled conditions to examine model performance and reproducibility.

1) Lookback Window Variation Experiment

- **Objective:** To determine how the length of past data references (from 10 to 100 days) affects prediction accuracy.
- **Method:** Vary the lookback window from 10 to 100 days in single-day increments, while holding all other hyperparameters (learning rate, dropout, batch size, random seed, etc.) constant.
- **Process:** Train each model for every window length and then measure classification accuracy (Accuracy) and the 95% confidence interval on the test set.
- **Significance:** In time-series forecasting, short-term features and long-term trends

can both be important. Finding an optimal lookback window may significantly affect model performance.

2) Random Seed Variation Experiment

- **Objective:** To verify how sensitive a model’s outcome is to stochastic factors such as weight initialization or mini-batch ordering, and thereby test whether claims of high accuracy are genuinely reproducible.
- **Method:** Fix the lookback window and other hyperparameters, but vary the random seed from 1 to 100. Each model is trained 100 times.
- **Process:** Analyze the accuracy and its 95% confidence interval for each seed, determining whether high accuracy is simply a chance result in specific cases or consistently repeatable.
- **Significance:** This addresses whether previously reported accuracies exceeding 70% are robust or heavily reliant on particular initial conditions.

Throughout both experiments, we maintained a chronological data split ($\approx 80\%$ for training and 20% for testing) to emulate real-world conditions in which future data cannot be known ahead of time. We also intermittently used a sliding-window retraining approach to prevent overfitting to particular time segments, ultimately examining performance across multiple yearly intervals.

2.2.3 Performance Metrics and Reproducibility Assessment

We employed **accuracy** (**Accuracy**) and the **95% confidence interval** (**CI**) to measure classification performance. From the confusion matrix (TP, TN, FP, FN), we calculated accuracy as:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}},$$

where TP, TN, FP, and FN respectively represent true positives, true negatives, false positives, and false negatives. For each experiment setting (lookback window or random seed), we derived a 95% confidence interval based on the distribution of accuracy values, assuming adequate sample size and approximate normality. This interval provides insight into the reliability of any given accuracy score and whether certain models maintain stable performance under different or repeated conditions.

2.2.4 Hyperparameter Stability

By analyzing the variance and standard deviation of the resulting accuracies, we can distinguish between models that occasionally achieve high scores under specific settings and models that demonstrate stable performance across diverse scenarios. Low variance combined with a high mean accuracy typically indicates lower sensitivity to hyperparameters or random seeds. Conversely, if the accuracy fluctuates widely with changes in the seed, one may question how practical such high accuracy would be in real-world financial markets.

These experimental designs enable a rigorous examination of whether claims ex-

ceeding 70% accuracy—often seen in existing research—are genuinely stable and reproducible. By systematically observing sensitivities to both the lookback window and random seed, we aim to shed light on the real-world applicability and inherent limitations of each model, including Transformers, in financial market forecasting.

Chapter 3

Results

This study conducted two main experiments to analyze the performance and reproducibility of financial time-series prediction models. The first experiment examined the effect of varying the lookback window (from 10 to 100 days) on prediction accuracy. The second experiment investigated how changes in the random seed (from 1 to 100) influenced accuracy outcomes. The models under evaluation were GRU, LSTM, and Transformer-based architectures. In this section, we present a quantitative analysis of the experimental results and discuss their implications.

3.1 Effect of Lookback Window on Performance

3.1.1 Experimental Setup

We varied the lookback window from 10 to 100 days in increments of one day, while keeping other hyperparameters (learning rate, batch size, dropout, etc.) constant. Each model was trained and tested using the same dataset split (80% training, 20% testing) and label definitions (Buy, Sell, Hold). The random seed was also fixed to ensure that only the lookback window differed across experiments.

3.1.2 Summary of Results

The accuracy distributions for GRU, LSTM, and Transformer models are shown in Figure 3.1 and Figure 3.2. The key statistical results are summarized in Table 3.1.

Table 3.1: Performance Metrics for Different Lookback Windows

Model	Max Accuracy	Min Accuracy	Mean Accuracy
GRU	0.6538	0.4048	0.5515
LSTM	0.6637	0.4667	0.5492
Transformer	0.6607	0.4618	0.5926

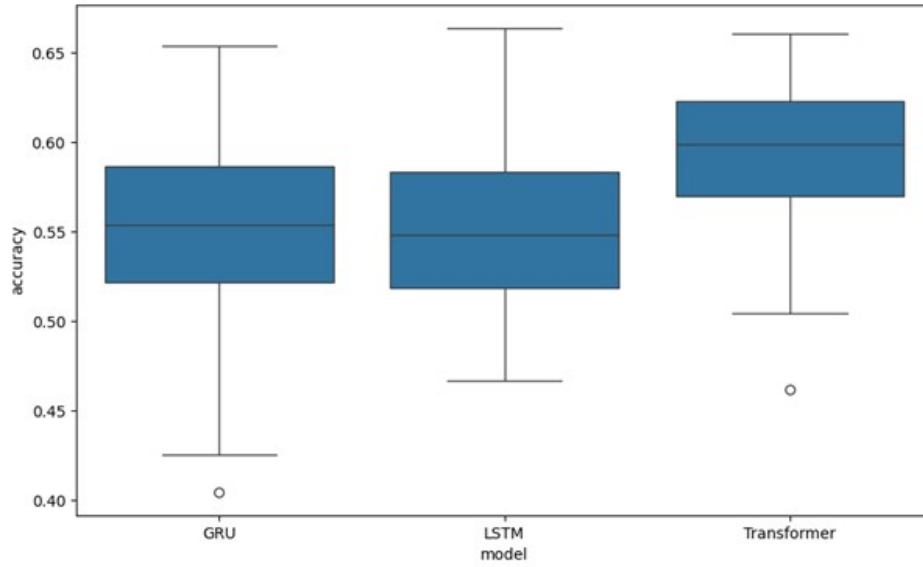


Figure 3.1: Boxplot of Accuracy Distribution across Lookback Windows.

3.1.3 Discussion

Key findings include:

- GRU and LSTM models exhibit strong sensitivity to lookback window selection, with accuracy peaking around 40-60 days and declining sharply outside this range.

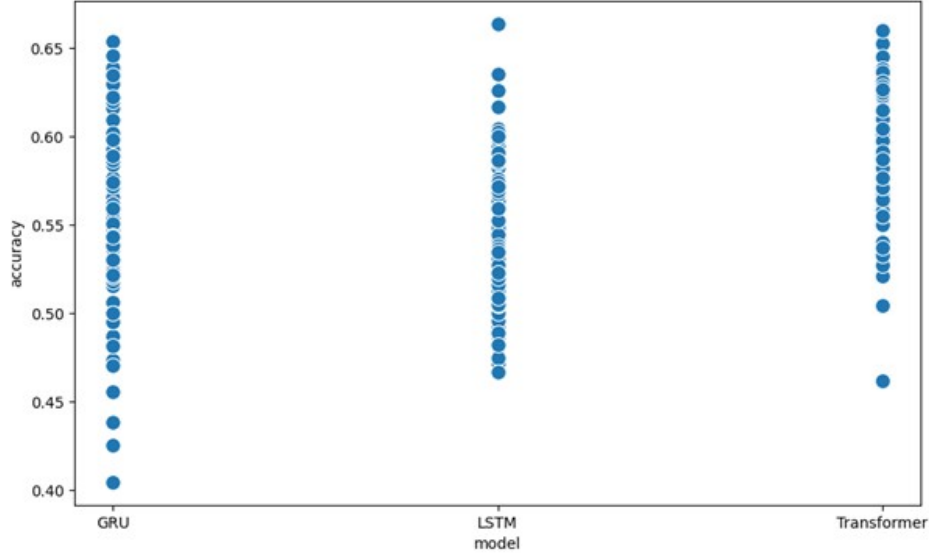


Figure 3.2: Scatter Plot of Accuracy Distribution across Lookback Windows.

- Transformer models showed higher mean accuracy (0.5926) and a more stable confidence interval (95% CI: 0.5849–0.6004).
- Excessively short or long lookback windows tend to reduce model performance, suggesting that an optimal window exists for capturing meaningful trends without introducing excessive noise.

3.2 Effect of Random Seed on Performance

3.2.1 Experimental Setup

In this experiment, we fixed the lookback window at 60 days and varied the random seed from 1 to 100. Each model was trained 100 times, maintaining identical hyperparameter settings (learning rate, batch size, dropout, etc.) and dataset splits.

3.2.2 Summary of Results

The accuracy distributions for each model under varying random seeds are depicted in Figure 3.3 and Figure 3.4. The key statistical results are summarized in Table 3.2.

Table 3.2: Performance Metrics for Different Random Seeds

Model	Max Accuracy	Min Accuracy	Mean Accuracy
GRU	0.6667	0.4924	0.5590
LSTM	0.6639	0.4677	0.5591
Transformer	0.6633	0.4683	0.5724

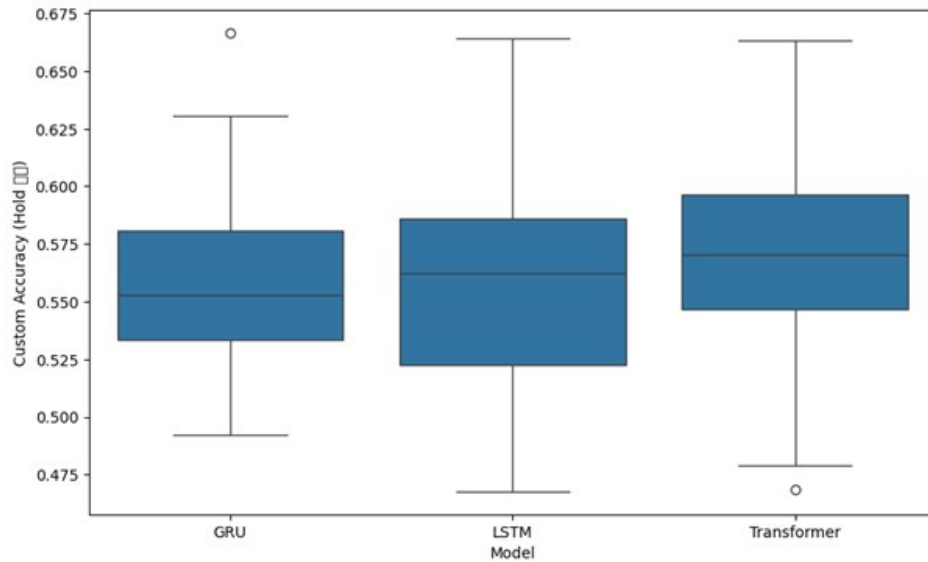


Figure 3.3: Boxplot of Accuracy Distribution across Different Seeds.

3.2.3 Discussion

Key findings from the random seed variation experiments include:

- Accuracy fluctuated significantly across different random seeds, with variations of up to ± 0.20 .

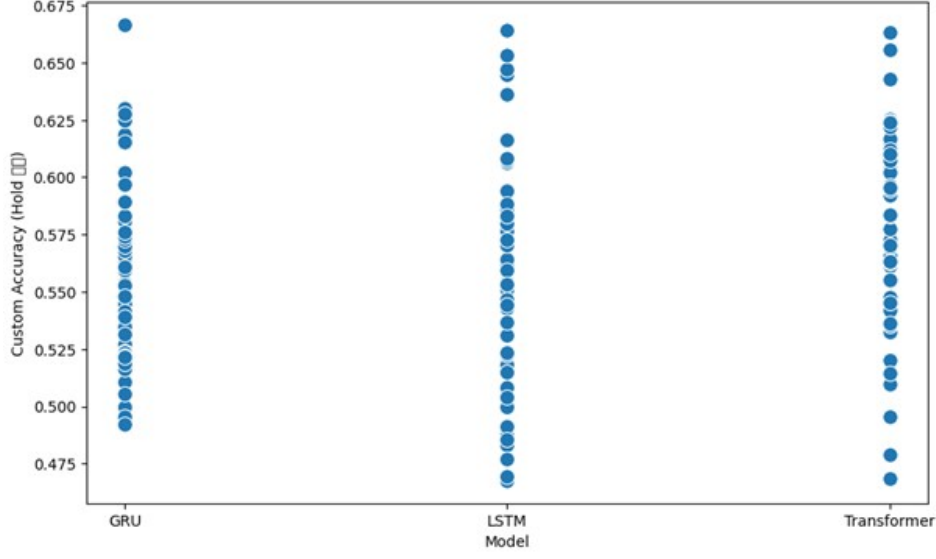


Figure 3.4: Scatter Plot of Accuracy Distribution across Different Seeds.

- GRU and LSTM models exhibited higher sensitivity to random initialization compared to the Transformer model.
- Transformer models had the highest mean accuracy (0.5724) and the narrowest confidence interval (95% CI: 0.5615–0.5834), indicating greater reproducibility.

3.3 General Interpretation: Balancing Overfitting and Reproducibility

Our findings suggest that the high accuracy values frequently reported in previous research may not always be reliable or reproducible. Key insights include:

- **High accuracy alone is misleading:** While all models reached peak accuracy around 0.66, their performance varied widely across different lookback windows and random seeds.
- **Mean accuracy and confidence intervals matter:** The Transformer model demonstrated the most stable performance, but RNN-based models achieved

comparable peak accuracy in specific cases.

- **Market conditions affect optimal settings:** Lookback windows of 40-60 days appeared optimal for capturing meaningful trends, but ideal settings may shift depending on market volatility.

By examining accuracy distributions instead of single-run results, this study highlights the importance of robustness and reproducibility when evaluating predictive models in financial markets.

Chapter 4

Conclusion

This study systematically examined the reproducibility and stability of deep learning models in financial time-series forecasting. Specifically, we compared traditional recurrent neural networks (GRU, LSTM) with Transformer-based models by analyzing their performance across two controlled experimental conditions: variations in lookback window length and changes in random seed initialization.

4.1 Summary of Findings

The experimental results demonstrated that GRU and LSTM models could achieve high accuracy values (65–66%) under specific conditions. However, these models exhibited significant fluctuations, with accuracy dropping sharply under different parameter settings. As a result, their mean accuracy converged to around 0.55, indicating inconsistent performance.

In contrast, the Transformer model achieved a maximum accuracy of 0.66, similar to the RNN models, but maintained a higher mean accuracy of approximately 0.59. Additionally, the 95% confidence interval (0.5615–0.5834) was narrower, suggesting greater stability and robustness in predictions.

4.2 Implications of Lookback Window and Random Seed Variations

The random seed variation experiments yielded similar trends to the lookback window experiments. Both GRU and LSTM models sometimes reached Transformer-level performance (66%) but were more sensitive to random initialization. Transformer models, on the other hand, demonstrated a higher mean accuracy (0.5724) and a narrower confidence interval, reinforcing their superiority in reproducibility.

These findings underscore a critical issue in financial forecasting: models that achieve exceptionally high accuracy in a single experimental setting may not necessarily generalize well to unseen market conditions. Instead, overfitting to specific parameter configurations (such as a particular lookback window or seed value) can lead to misleadingly optimistic results.

4.3 Key Insights and Future Directions

The primary takeaway from this research is that evaluating model performance based solely on peak accuracy is insufficient. Instead, assessing mean performance, variance, and robustness across multiple experimental conditions (lookback windows, seeds, hyperparameters) is crucial.

Given the complex and non-stationary nature of financial markets, the following directions should be prioritized in future research:

- Enhancing generalizability: Implementing cross-validation techniques or expanding the use of sliding window methods to ensure robust model evaluation.
- Beyond accuracy metrics: Incorporating real-world trading simulations to mea-

sure profitability, maximum drawdown (MDD), and transaction costs.

- Adaptive modeling strategies: Developing hybrid approaches that dynamically adjust lookback periods and optimize model performance based on market conditions.

Ultimately, this study highlights the importance of reproducibility and stability in financial forecasting models. While Transformer models demonstrated superior performance consistency, the variability observed in RNN-based models suggests that certain architectures may still perform well in specific market conditions. Future work should focus on bridging the gap between theoretical accuracy and practical trading performance, ensuring that AI-driven financial models are not only statistically sound but also viable in real-world applications.

Summary

Revisiting Bitcoin Price Prediction with On-Chain and Financial Data:

A Reflection on Reproducibility

This study investigates the reproducibility and stability of deep learning models in financial time-series forecasting, specifically focusing on Bitcoin price prediction. While prior research has explored the application of LSTM, GRU, and Transformer-based models for predicting asset prices, their reproducibility across different settings has often been overlooked. This study systematically analyzes model performance variations by altering two critical factors: (1) the lookback window (ranging from 10 to 100 days) and (2) the random seed (varying from 1 to 100).

Our findings indicate that although GRU and LSTM models occasionally achieve high accuracy levels (up to 66%), they exhibit substantial performance fluctuations, with an average accuracy stabilizing around 0.55. Conversely, the Transformer model, despite achieving a similar maximum accuracy, demonstrated a more stable performance with a higher mean accuracy of approximately 0.59 and a narrower confidence interval. This suggests that the Transformer model is relatively more robust under varying experimental conditions.

Furthermore, our results highlight the limitations of relying solely on peak accuracy values to evaluate model efficacy. We observed that high accuracy can often be a result of specific hyperparameter configurations or favorable initial conditions rather than

a genuinely robust predictive capability. The analysis of lookback window variation suggests that financial markets exhibit both short-term fluctuations and long-term trends, making it essential to carefully select an optimal window size. Similarly, the sensitivity to random seed variation reveals that deep learning models can produce widely different predictions based on seemingly minor changes in initialization settings.

The key takeaway from this research is that high accuracy claims in financial deep learning models must be interpreted cautiously. Instead of emphasizing a single best-performing setting, researchers and practitioners should assess average performance, variance, and reproducibility across multiple configurations. Future research should extend this framework by incorporating additional evaluation metrics such as trading profitability, maximum drawdown (MDD), and transaction costs to better assess the real-world applicability of AI-driven financial forecasting models.

References

1. A. Chennupati, B. D. Jasvitha, B. Prahas, M. K., and B. A. Ghali, “Comparative analysis of bitcoin price prediction models: Lstm, bilstm, arima and transformers,” in *Proc. 15th Int. Conf. Comput. Commun. Netw. Technol. (ICCCNT)*, (IIT Mandi, India), 2024.
2. G. Kim, D.-H. Shin, J. G. Choi, and S. Lim, “A deep learning-based cryptocurrency price prediction model that uses on-chain data,” *IEEE Access*, vol. 10, pp. 56232–56235, Jun. 2022.
3. I. E. Livieris, N. Kiriakidou, S. Stavroyiannis, and P. Pintelas, “An advanced cnn-lstm model for cryptocurrency forecasting,” *Electronics*, vol. 10, p. 287, Jan. 2021.
4. B. Yildirim and M. Taskiran, “Optuna based optimized transformer model approach in bitcoin time series analysis,” in *Proc. 26th Int. Conf. Digit. Signal Process. Appl. (DSPA)*, (Istanbul, Turkey), 2024.
5. A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Proc. 31st Adv. Neural Inf. Process. Syst. (NeurIPS)*, pp. 5998–6008, 2017.
6. Y. Han, “Bitcoin transformer implementation code.” <https://github.com/your-username/bitcoin-transformer>, 2025. Accessed: 2025-06-11.